

MAC 2312 — CALCULUS II

Lecture 5: Arc Length

Monday, September 14 • Sections 2.4

• Guided Notes •

How to Use These Guided Notes: These guided notes are designed to help you organize the main ideas from each lecture but

1. we will not complete every example in class.
2. Although most definitions and theorems are included, you may need to add additional notes, examples, or explanations during class.
3. Occasionally, you will see sections labeled **Deeper Insight**. These sections provide interesting background information and help develop a deeper understanding of calculus, but they are not part of the core Calculus II content.

Our final application of integration in this unit is the calculation of arc length. We will revisit arc length later in the course when studying other representations of curves.

Theorem 5.1 – Arc length - Integrating with respect to x

Suppose f' is continuous on $[a, b]$. Then, the length L of the portion of the curve $y = f(x)$ from $x = a$ to $x = b$ is

$$L = \int_a^b \sqrt{1 + (f'(x))^2} dx.$$

Example 5.1

We all know that the perimeter of the circle with radius R is $2\pi R$.
Let us use the arc length formula to verify this result.

1. Draw a circle with radius R and center at $(0, 0)$.
2. What relation satisfy the points (x, y) on the circle? _____
3. Can the entire circle be represented by one equation of the form $y = f(x)$? Explain. Then express the upper semicircle as $y = f(x)$.

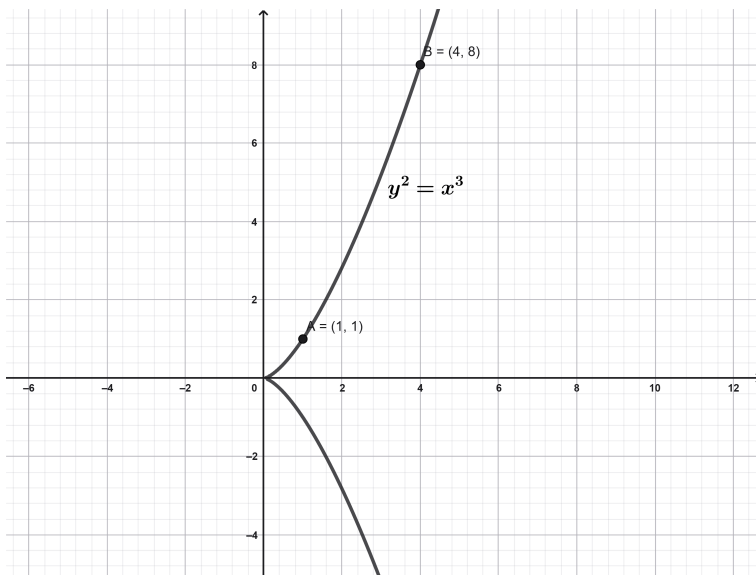
4. Set up a definite integral equal to the perimeter of the circle. _____

5. Evaluate the resulting definite integral. You may use that

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin(x) + C.$$

Example 5.2

Find the length of the semicubical parabola $y^2 = x^3$ between the points $(1, 1)$ and $(4, 8)$.



1. Write the points on the curve as $y = f(x)$: _____
2. Set up the definite integral equal to the arc length: _____
3. Evaluate the definite integral.

A curve may be easier to describe as $x = g(y)$ than as $y = f(x)$. In that case, we can calculate its length by integrating with respect to y .

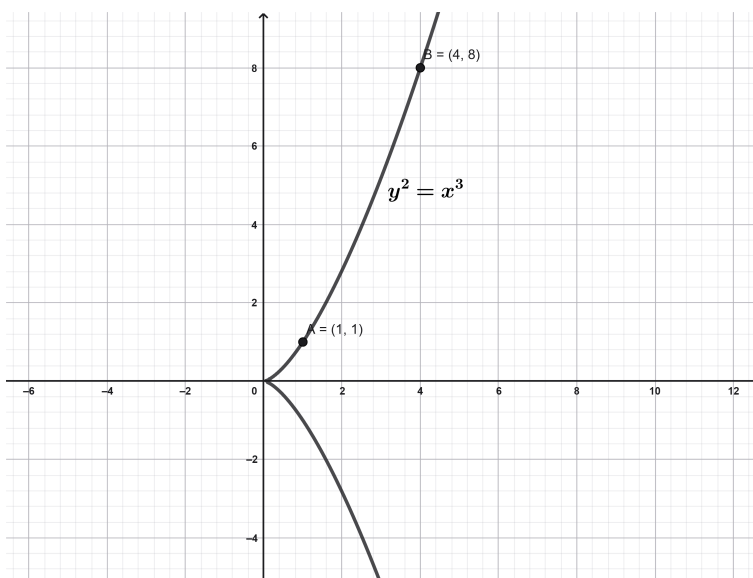
Theorem 5.2 – Arc length - Integrating with respect to y

Suppose g' is continuous on $[c, d]$. Then, the length L of the portion of the curve $x = g(y)$ from $y = c$ to $y = d$ is

$$L = \int_c^d \sqrt{1 + (g'(y))^2} dy.$$

Example 5.3

Set up the definite integral to determine the arc length we calculated in the previous example, but integrating with respect to y .



1. Write the points on the curve as $x = g(y)$: _____
2. Set up the definite integral equal to the arc length: _____

In general, arc length may be calculated by integrating with respect to x or with respect to y :

- To integrate with respect to x , describe the curve as $y = f(x)$ and use x -values as the limits of integration.
- To integrate with respect to y , describe the curve as $x = g(y)$ and use y -values as the limits of integration.

When both representations are possible, choose the one that produces the simpler derivative and integrand.

Try it yourself. Find the exact length of the curve

$$y = \frac{1}{2} \ln(\sin(2x)), \quad \frac{\pi}{8} \leq x \leq \frac{\pi}{6}.$$

1. Simplify $1 + (f'(x))^2$ with $f(x) = \frac{1}{2} \ln(\sin(2x))$. Use the following identities

$$\cot(x) = \frac{\cos(x)}{\sin(x)} \quad \text{and} \quad 1 + \cot^2(x) = \csc^2(x).$$

2. Why may we simplify the definite integral equal to the arc length to

$$\int_{\frac{\pi}{8}}^{\frac{\pi}{6}} \csc(2x) \, dx ?$$

3. Evaluate the definite integral. You may use that

$$\int \csc(x) \, dx = \ln |\csc(x) - \cot(x)| + C.$$

Now, solve some practice problems. Try to avoid the following common errors.

! Common Mistakes

- Using $f(x)$ instead of $f'(x)$ in the arc-length formula.
- Forgetting the 1 or forgetting to square the derivative in

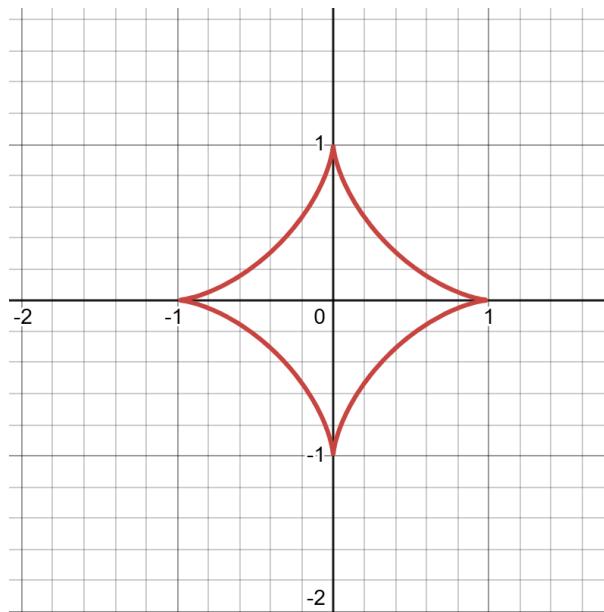
$$\sqrt{1 + (f'(x))^2}.$$

- Writing $\sqrt{u^2} = u$ instead of $\sqrt{u^2} = |u|$.

Practice Problems

Level I problems

1. Find the length of an astroid $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 1$.



2. Set up an integral which represents the arc length of the curve $f(x) = (x^4 - 1)^{\frac{3}{2}}$ from $x = 1$ to $x = 3$.
3. Set up an integral which represents the arc length of the curve $f(x) = (x^3 - 1)^{\frac{3}{2}}$ from $x = 1$ to $x = 4$.
4. Find the exact length of the curve $y = \ln(\cos(x))$, $0 \leq x \leq \frac{\pi}{3}$.
5. Find the length of the curve $x = e^y + \frac{1}{4}e^{-y}$, $0 \leq y \leq 1$.

By the end of today you will be able to:

- Recognize the geometric structure of the arc-length formula.
- Choose whether to integrate with respect to x or y and set up the corresponding arc-length integral.
- Simplify and evaluate an arc-length integral when the expression under the square root becomes a perfect square.

Muddiest Point: What concept from today's lesson is still unclear?